

	Work Instruction	Doc. Revision	01
	AXI V810 STD X-Axis MCG Servo Motor Removal and Replacement Procedure	Effective Date	2 Oct 2016

This procedure is applicable to below models:

V810 STD	☒	S1	☒	S2
V810 XXL	☐	S1	☐	S2
V810 S2EX	☐			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	09 Dec 2015	First Release	CHIN CHEE YAN
01	2 Oct 2016	Template Changes & Content Update	ONG KEAN SHENG

	Work Instruction	Doc. Revision	01
	AXI V810 STD X-Axis MCG Servo Motor Removal and Replacement Procedure	Effective Date	2 Oct 2016

Content

X-Axis MCG Servo Motor Removal.....	3
X-Axis MCG Servo Motor Replacement.....	5

	Work Instruction	Doc. Revision	01
	AXI V810 STD X-Axis MCG Servo Motor Removal and Replacement Procedure	Effective Date	2 Oct 2016

X-Axis MCG Servo Motor Removal

1. Unload if any board inside machine.
2. Power off the SSCA main power # Importance.
3. Open all front access door for ease of removal process.
4. Manually position the stage to desire position for ease of removal process.
5. Disconnect the connector from V810 Cable, Motor ENC/Hall, X Axis (P/N, 4XASC-A074) and V810 Cable, Motor PWR, X Axis (P/N, 4XASC-A075) as shown. Use flat head screwdriver to unscrew the lock screws as below Figure 1.

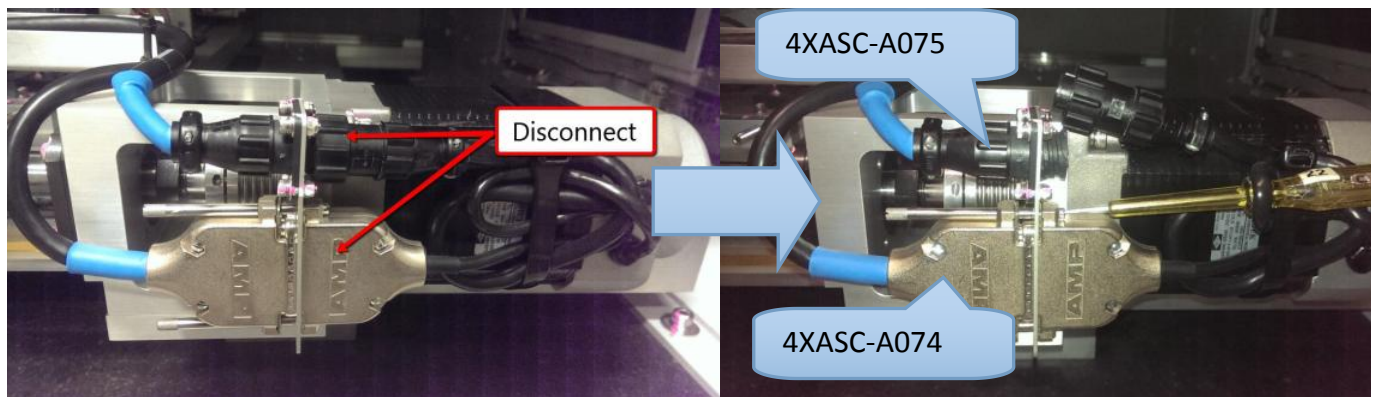


Figure 1 MCG Motor connector removal

6. Loosen the 1x screw at Bellow Coupling_Bore Dia: 12.7mm/12.7mm (P/N, 2660-0032) as shown.

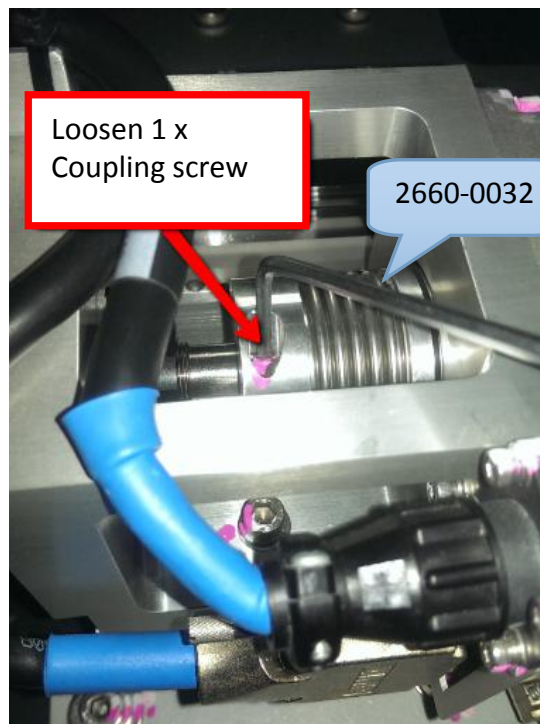


Figure 2 Coupling Screw removal

ViTrox	Work Instruction	Doc. Revision	01
	AXI V810 STD X-Axis MCG Servo Motor Removal and Replacement Procedure	Effective Date	2 Oct 2016

7. Loosen 4 x screws at HOUSING, MOTOR X-AXIS (P/N, 4XAXY-010) as shown.

Remarks : Require extra length Allen wrench (M5) to remove inner side screw.

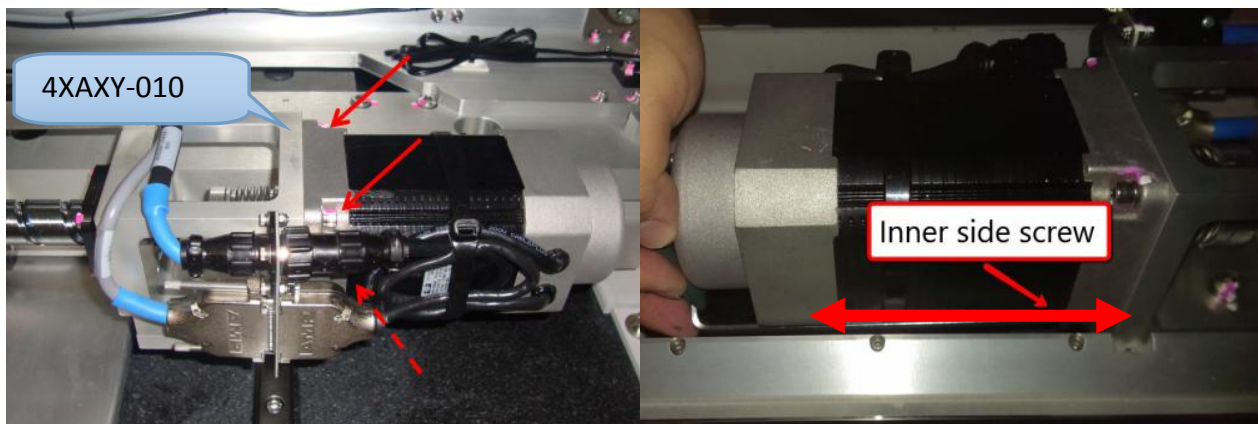


Figure 3 MCG servo motor removal

8. Gently remove the motor with coupling from 4XAXY-010 (Motor Housing)

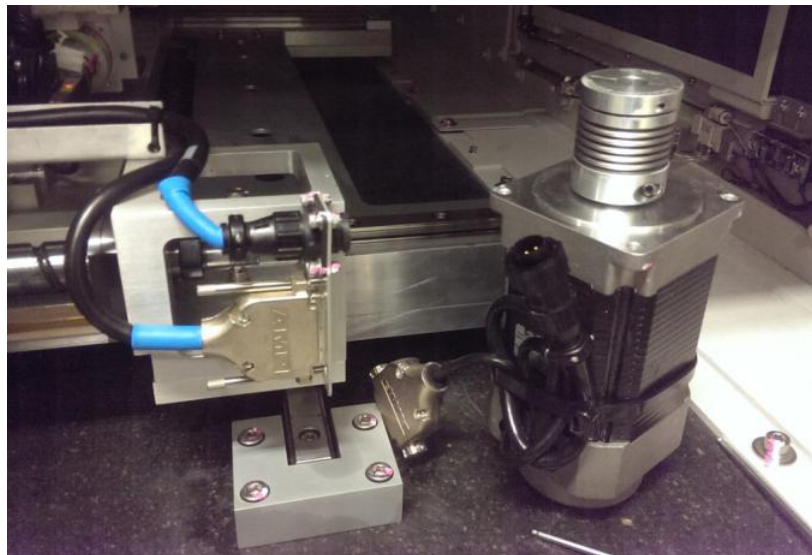


Figure 4 MCG servo motor removal

9. Loosen 1 x screw at other end of 2660-0032(Coupling) and remove accordingly.

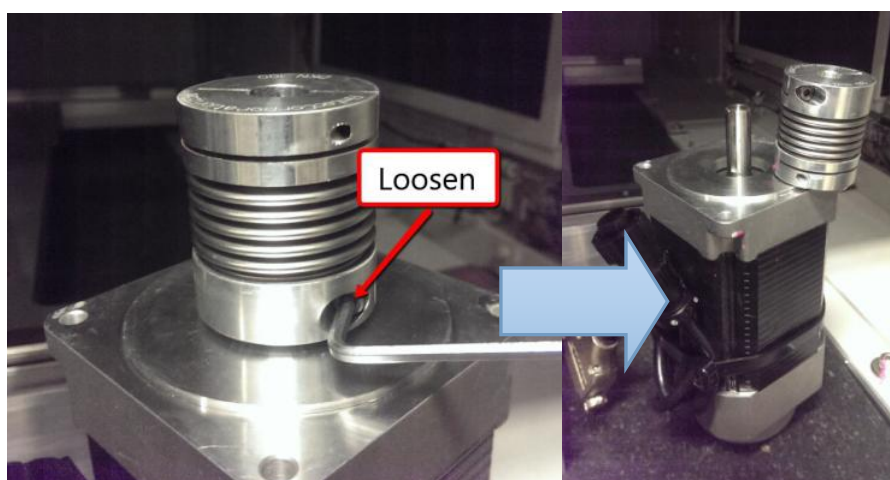


Figure 5 Coupling removal

	Work Instruction	Doc. Revision	01
	AXI V810 STD X-Axis MCG Servo Motor Removal and Replacement Procedure	Effective Date	2 Oct 2016

X-Axis MCG Servo Motor Replacement

10. Prepare new V810 MOTOR, BRLSS DC SERVO ENC5000 (P/N, 4XAXY-049) as shown.

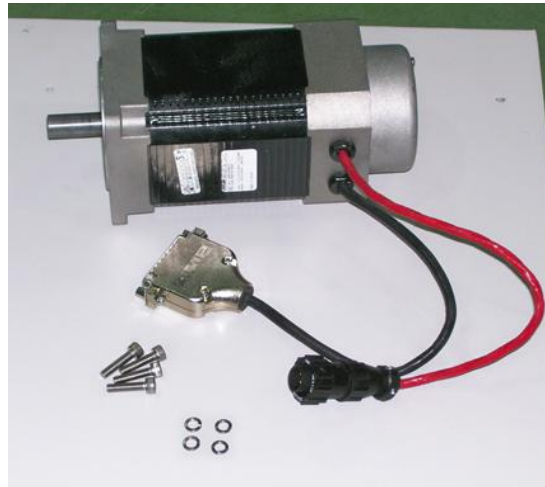


Figure 6 MCG servo motor

11. Assemble 2660-0032 (Coupling) to V810 X-Axis Ball screw Kit (P/N, 4XASP-A024) and tighten 1 x screw only at coupling (End shaft of X axis Ball Screw).

Special remarks :

Both screws at coupling require to be visible

Socket head cap screw could not worn out

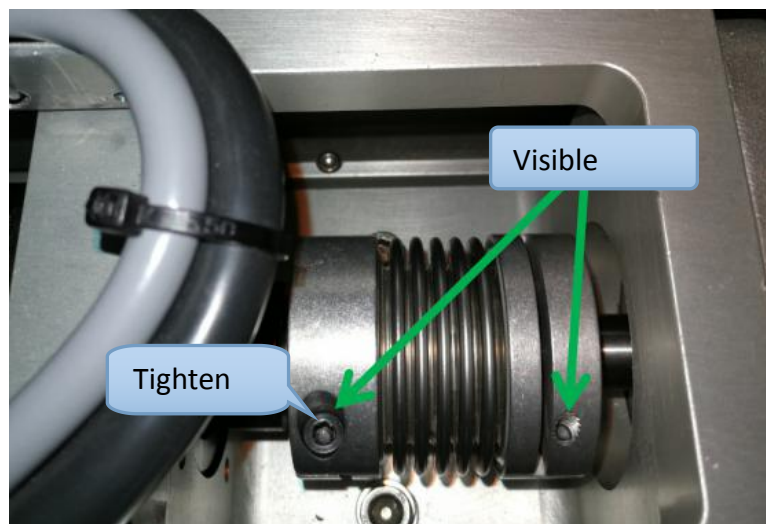


Figure 7 Coupling fastening

ViTrox 	Work Instruction	Doc. Revision	01
	AXI V810 STD X-Axis MCG Servo Motor Removal and Replacement Procedure	Effective Date	2 Oct 2016

12. Assemble 2670-0035 (Teknic Motor) to 4XAXY-010 (Motor Housing) as shown & tighten up 4 x screws.

Special remarks :

Socket head cap screw could not worn out



Figure 8 Servo motor assembly

13. Tighten screw at the other end of 2660-0032 (Coupling) as shown.

Special remarks :

Socket head cap screw could not worn out

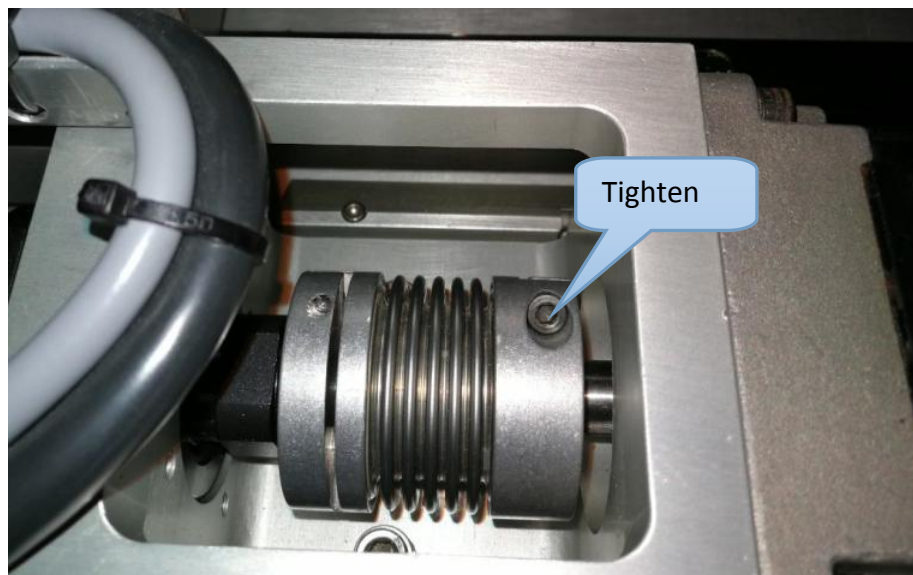


Figure 9 Coupling screw fastening

ViTrox	Work Instruction	Doc. Revision	01
	AXI V810 STD X-Axis MCG Servo Motor Removal and Replacement Procedure	Effective Date	2 Oct 2016

14. Route the cable as shown and apply cable to tidy up the extra length cable. May refer to below Figure 10.

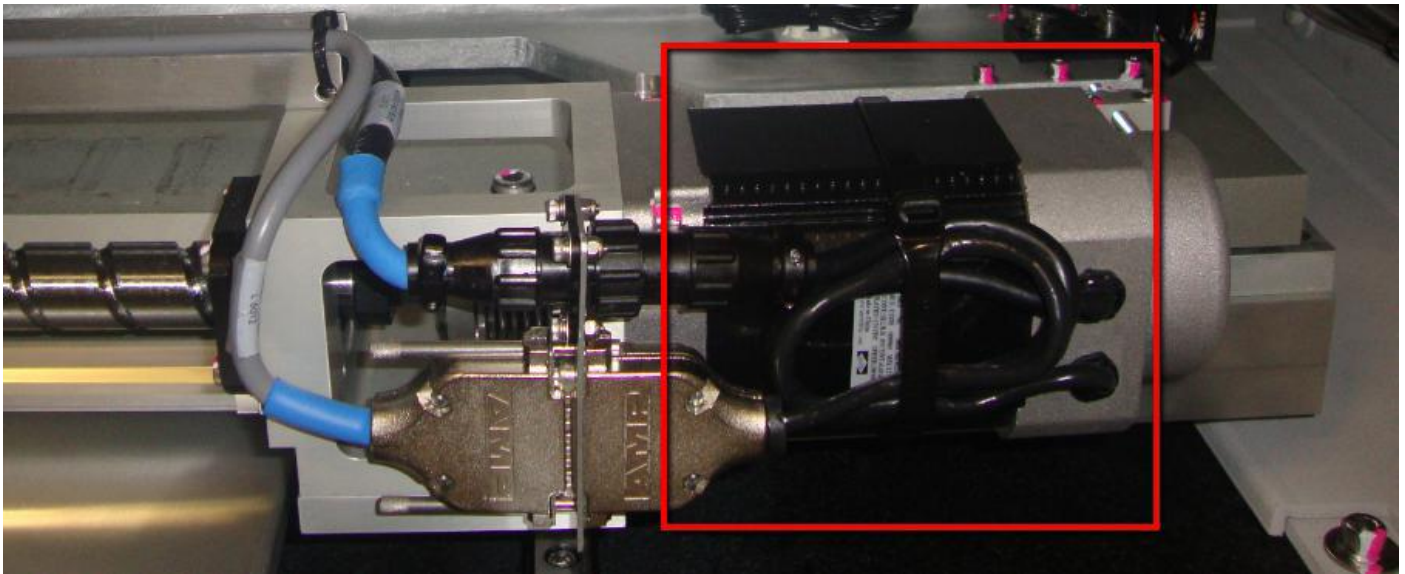


Figure 10 Motor cable routing [COMPLETE]

15. Connect 4XAXY-049 (MCG Servo Motor) to 4XASC-A074 (Rotary encoder) and 4XASC-A075 (Power Connector) as shown. Tighten the lock screws as well.

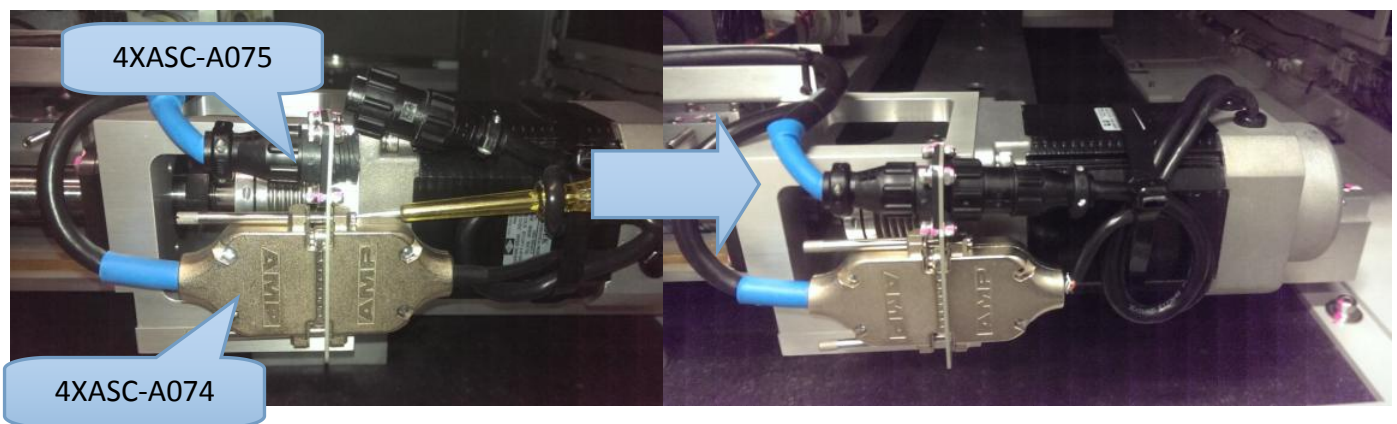


Figure 11 Power Connector & Rotary Encoder

16. Remove all tools from machine stage.
17. Ensure all screws are fully tightened.
18. Manually exercise X axis stage in “Front to Rear “or “Rear to Front” for few routes. . If alignment is require, may refer to procedure **“AXI V810 STD X-Axis Ball Screw Alignment Procedure”**
19. Close all the access door.
20. Go to Panel Positional, load Long scan.path and run 10 loops looping test.
21. Run 1 loop of CD & A.